

Project Design Phase-II

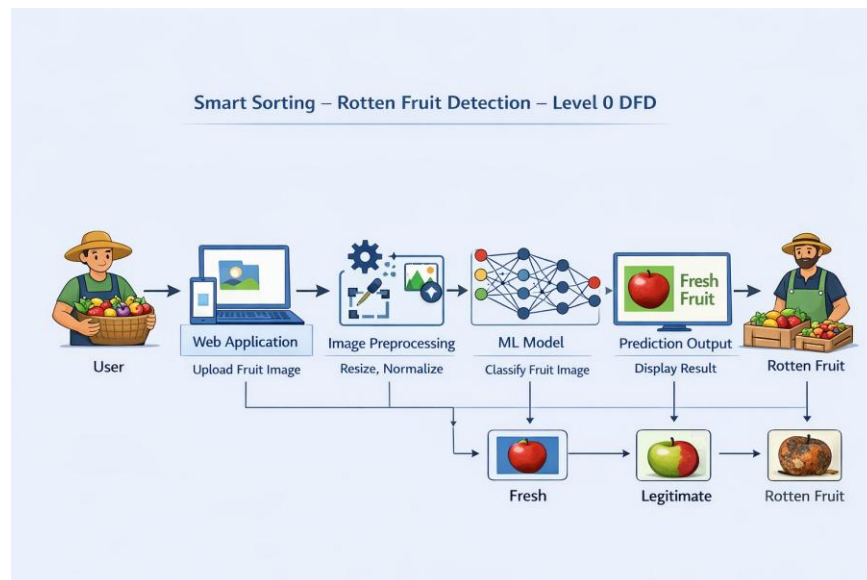
Data Flow Diagram & User Stories

Parameter	Details
Date	12 February 2026
Team ID	LTVIP2026TMIDS79181
Project Name	Smart Sorting – Rotten Fruit Detection using Machine Learning
Maximum Marks	4 Marks

Data Flow Diagram (DFD - Level 0)

A **Level-0 Data Flow Diagram** illustrates how data flows between the user, web application, image preprocessing module, machine learning model, and prediction output system.

Data Flow:



User Stories

User Type	Epic	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Fruit Vendor	Fruit Quality Detection	USN-1	As a vendor, I can upload fruit images to check	System correctly classifies fruit	High	Sprint-1

			whether the fruit is fresh or rotten.	image as Fresh or Rotten.		
Farmer	Quality Monitoring	USN-2	As a farmer, I can detect rotten fruits quickly before selling them in the market.	Rotten fruits are identified accurately by the system.	High	Sprint-1
Quality Inspector	Monitoring	USN-3	As an inspector, I can verify fruit quality using the system.	System displays prediction results with confidence score.	High	Sprint-2
Market Manager	Reporting	USN-4	As a manager, I can monitor fruit quality statistics in batches.	System generates reports on fresh vs rotten fruit detection.	Medium	Sprint-2
System Admin	Model Management	USN-5	As an admin, I can update or retrain the machine learning model when new data is available.	Updated model is deployed without affecting application performance.	Medium	Sprint-3